

IMPLEMENTATION AND VALIDATION OF THE $k-\omega_0$ TURBULENCE MODEL

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ABSTRACT

The standard $k-\omega$ model employs a singular solution to the ω equation, which poses challenges for numerical discretisation. A reformulation using a transformation maps a nonsingular regular solution ω_0 into the singular solution, improving numerical robustness while retaining the behaviour of the standard model. In the present study, we implement and verify both the standard $k-\omega$ and the reformulated $k-\omega_0$ models. The $k-\omega_0$ formulation is shown to yield better-converged results on the same grid with reduced numerical stiffness. The models are further assessed for flow over a periodic hill and high-speed boundary layers, demonstrating the improvements offered by the reformulated approach.

NOMENCLATURE

$\bar{A}$	Reynolds/time-averaged quantity
$\tilde{A}$	Favre-averaged quantity
ξ, η, ζ	Generalized curvilinear coordinates
α, β_0	Closure coefficients

1 INTRODUCTION

Turbulence modelling remains a cornerstone of computational fluid dynamics, with two-equation models being widely used in engineering applications. Among these, the $k-\omega$ model has been a popular choice for wall-bounded flows [1]. However, the standard $k-\omega$ model employs a singular solution to the ω equation near the wall. Asymptotically,

$$\omega \rightarrow \frac{6\nu}{\beta_0 y^2} \quad (1)$$

as $y \rightarrow 0$. This singular behaviour is a source of difficulty in numerical implementations [2].

Thus, the $k-\omega$ model requires careful treatment of boundary conditions, and most practical approaches rely on strategies that introduce arbitrary constants. This has motivated efforts to reformulate the ω equation in a more computationally robust form. Recently, Durbin and Yin [3] introduced a transformation that maps a nonsingular solution ω_0 into the singular one, showing improved behaviour in both low-speed and high-Mach-number flows. The present work follows this formulation to implement the $k-\omega_0$ model. The models are first verified against flat plate boundary layer flow, after which their convergence and numerical stiffness are compared on the same grid. The $k-\omega_0$ model is then applied to flow over a periodic hill and high-speed boundary layers to examine the practical benefits of the reformulation.

2 NUMERICAL FRAMEWORK

The turbulent flat-plate boundary-layer flow was simulated at a Mach number of 0.2 and a Reynolds number of 5×10^6 . Figure 1 illustrates the computational domain and the corresponding boundary conditions. A structured cartesian mesh consisting of 69×49 grid points was chosen to satisfy the available computational resources and time constraints. The results obtained were validated against reference data of NASA's compressible RANS solver CFL3D. This data was generated on a significantly finer grid comprising 545×385 points and was provided on the Turbulence Modelling Resource (TMR) database [4].

2.1 Governing Equations - Standard $k-\omega$ Model

The present work solves the compressible Reynolds-Averaged Navier-Stokes (RANS) equations with Favre averaging.

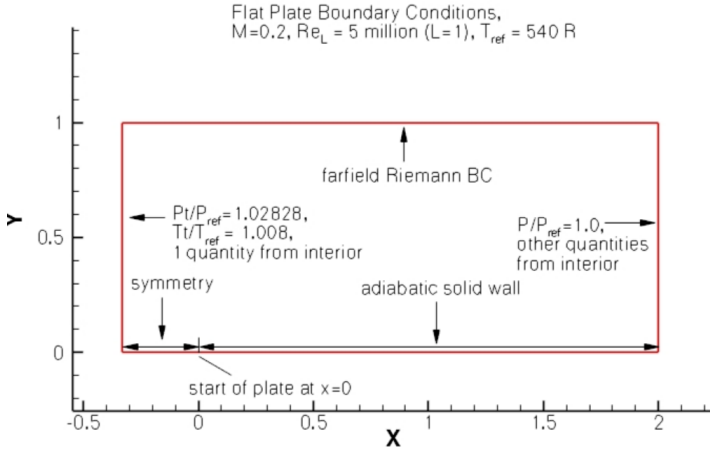


FIGURE 1: Computational Domain and Boundary Conditions [4]

Turbulence closure is provided by the two-equation $k-\omega$ model (Wilcox2006-klim-m) [1, 4], which introduces additional transport equations for the turbulent kinetic energy k and the specific dissipation rate ω . The governing equations are as follows:

Mass Conservation:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_i)}{\partial x_i} = 0 \quad (2)$$

Momentum Conservation:

$$\frac{\partial (\bar{\rho} \tilde{u}_i)}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_j \tilde{u}_i)}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} + \frac{\partial}{\partial x_j} (\bar{\rho} \tau_{ij}) \quad (3)$$

Energy Conservation:

$$\begin{aligned} \frac{\partial (\bar{\rho} (\tilde{e} + \frac{1}{2} \tilde{u}_i \tilde{u}_i + k))}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_j (\tilde{h} + \frac{1}{2} \tilde{u}_i \tilde{u}_i + k))}{\partial x_j} = \\ \frac{\partial}{\partial x_j} \left[\left(\frac{\mu}{Re_\infty Pr_L} + \frac{\mu_T}{Re_\infty Pr_T} \right) \frac{\partial \tilde{h}}{\partial x_j} + \left(\frac{\mu}{Re_\infty} + \sigma^* \frac{\bar{\rho} k}{\omega} \right) \frac{\partial k}{\partial x_j} \right] \\ + \frac{\partial}{\partial x_j} [\tilde{u}_i (\bar{\tau}_{ij} + \bar{\rho} \tau_{ij})] \end{aligned} \quad (4)$$

Turbulent Kinetic Energy:

$$\begin{aligned} \frac{\partial (\bar{\rho} k)}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_j k)}{\partial x_j} = \bar{\rho} \tau_{ij} \frac{\partial (\tilde{u}_i)}{\partial x_j} - \beta^* \bar{\rho} k \omega \\ + \frac{\partial}{\partial x_j} \left[\left(\frac{\mu}{Re_\infty} + \sigma^* \frac{\bar{\rho} k}{\omega} \right) \frac{\partial k}{\partial x_j} \right] \end{aligned} \quad (5)$$

Specific Dissipation Rate:

$$\begin{aligned} \frac{\partial (\bar{\rho} \omega)}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_j \omega)}{\partial x_j} = \alpha \frac{\omega}{k} \bar{\rho} \tau_{ij} \frac{\partial \tilde{u}_i}{\partial x_j} - \beta \bar{\rho} \omega^2 + \sigma_d \frac{\bar{\rho}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \\ + \frac{\partial}{\partial x_j} \left[\left(\frac{\mu}{Re_\infty} + \sigma^* \frac{\bar{\rho} k}{\omega} \right) \frac{\partial \omega}{\partial x_j} \right] \end{aligned} \quad (6)$$

The molecular stress tensor ($\bar{\tau}_{ij}$) and Reynolds stress tensors (τ_{ij}) are given by:

$$\begin{aligned} \bar{\tau}_{ij} = 2 \frac{\mu}{Re_\infty} \bar{S}_{ij}, \quad \bar{\rho} \tau_{ij} = 2 \frac{\mu_T}{Re_\infty} \bar{S}_{ij} - \frac{2}{3} \bar{\rho} k \delta_{ij} \\ \bar{S}_{ij} = S_{ij} - \frac{1}{3} \frac{\partial \tilde{u}_k}{\partial x_k} \delta_{ij} \end{aligned} \quad (7)$$

Here, S_{ij} is the strain-rate tensor. The turbulent eddy viscosity (μ_T) is found as follows

$$\mu_T = \frac{\bar{\rho} k}{\tilde{\omega}}, \quad \tilde{\omega} = \max \left\{ \omega, C_{lim} \sqrt{\frac{2 \bar{S}_{ij} \bar{S}_{ij}}{\beta^*}} \right\}, \quad C_{lim} = \frac{7}{8} \quad (8)$$

The closure coefficients $\alpha, \beta_0, \beta^*, \sigma, \sigma^*, \sigma_{d0}$ and β_0 are 13/25, 9/100, 1/2, 3/5, 1/8 and 0.0708 respectively. β is found from β_0 as $\beta_0 f_\beta$, where f_β is given by

$$f_\beta = \frac{1 + 85 \chi_\omega}{1 + 100 \chi_\omega}, \quad \chi_\omega = \left| \frac{\Omega_{ij} \Omega_{jk} \hat{S}_{ki}}{(\beta^* \omega)^3} \right|, \quad \hat{S}_{ki} = S_{ki} - \frac{1}{2} \frac{\partial \tilde{u}_m}{\partial x_m} \delta_{ki} \quad (9)$$

where $\Omega_{ij} = \frac{1}{2} \left(\frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{u}_j}{\partial x_i} \right)$. The constant σ_d found in the cross-diffusion term of the ω -equation, is modelled as

$$\sigma_d = \begin{cases} 0, & \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \leq 0 \\ \sigma_{d0}, & \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} > 0 \end{cases} \quad (10)$$

Furthermore, the production term appearing in the k and ω equations, $P_k = \bar{\rho} \tau_{ij} \frac{\partial \tilde{u}_i}{\partial x_j}$ is approximated as $P_k = \mu_T S^2$, where $S = \sqrt{2 S_{ij} S_{ij}}$. The production is limited by $P_k = \min(P_k, \beta^* \bar{\rho} \omega k)$

2.2 Reformulation to the $k-\omega_0$ Model

To handle the numerical stiffness introduced by the wall boundary condition of the standard $k-\omega$ model, the proposed method

is to solve for a regular variable ω_0 . [3] The relationship between the actual singular solution(ω) and the regular solution(ω_0) is governed by the transformation:

$$\omega = \frac{6\nu}{\beta_0 y^2} F\left(\frac{\omega_0 y^2}{\nu}\right) \quad (11)$$

where $F(X)$ is given by:

$$F(X) = \frac{\beta_0}{6} X + e^{-AX} \quad (12)$$

Here, A is an empirical constant, found to be $A = 0.012$ by Durbin and Yin [3]. The transformation function $F(X)$ satisfies two asymptotic constraints: at the wall, $F(0) = 1$, recovering the exact singular asymptote $\omega \rightarrow \frac{6\nu}{\beta_0 y^2}$, and far from the wall, $F(X) \rightarrow \frac{\beta_0 X}{6}$, ensuring that $\omega \rightarrow \omega_0$.

Modified Transport Equation:

The standard transport equations are solved, but the variable ω is replaced by the regular variable ω_0 in its transport equation. Equation (5) remains the same however the singular ω in Equation (6) is replaced by the regular ω_0 :

$$\begin{aligned} \frac{\partial(\bar{\rho}\omega_0)}{\partial t} + \frac{\partial(\bar{\rho}\tilde{u}_j\omega_0)}{\partial x_j} &= \alpha \frac{\omega_0}{k} \bar{\rho} \tau_{ij} \frac{\partial \tilde{u}_i}{\partial x_j} - \beta \bar{\rho} \omega_0^2 + \sigma_d \frac{\bar{\rho}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega_0}{\partial x_j} \\ &+ \frac{\partial}{\partial x_j} \left[\left(\frac{\mu}{Re_\infty} + \sigma \frac{\bar{\rho}k}{\omega} \right) \frac{\partial \omega_0}{\partial x_j} \right] \end{aligned} \quad (13)$$

The singular solution obtained by the transformation of the regular variable is used to compute the turbulent eddy viscosity:

$$\nu_t = \frac{k}{\omega} \quad (14)$$

Boundary Conditions for ω_0 :

Two wall boundary conditions can be applied for ω_0 :

1. $\omega_0(0) = 0$
2. $\omega_0(0) = \sqrt{\frac{\alpha}{\beta_0}} \frac{\tau_w}{\mu}$

The second boundary condition is particularly advantageous for numerical solvers. In the series expansion of ω_0 , the quadratic term takes the form $(2C_{\omega_1}|S_0|^2 - C_{\omega_2}\omega_{00}^2)y^2/2\nu$. Applying $\omega_{00} = \sqrt{C_{\omega_1}/C_{\omega_2}} \tau_w/\mu$ causes this term to vanish identically, making ω_0 nearly constant for $y^+ < 7$ [3].

2.3 Spatial Discretisation and Filtering

The spatial discretisation of interior points follows the compact finite-difference stencil [5]:

$$\alpha \phi'_{i-1} + \phi'_i + \alpha \phi'_{i+1} = b \frac{\phi_{i+2} - \phi_{i-2}}{4} + a \frac{\phi_{i+1} - \phi_{i-1}}{2} \quad (15)$$

where $\alpha = 0$ recovers an explicit scheme, while $\alpha \neq 0$ yields a compact (implicit) scheme with improved spectral resolution. At non-periodic boundaries, one-sided stencils of the form

$$\sum_k \alpha_k \phi'_k = \sum_j a_j \phi_j \quad (16)$$

are applied, where the coefficients α_k and a_j are adopted from Gaitonde and Visbal (1998) [5]. The boundary stencils involve up to seven grid points and are applied symmetrically at both ends of the domain.

High-order central difference schemes, while accurate, are susceptible to spurious high-frequency oscillations. To suppress these instabilities without compromising solution accuracy, implicit spectral filters were applied to introduce numerical dissipation at high wavenumbers.

$$\alpha_f \hat{\phi}_{i-1} + \hat{\phi}_i + \alpha_f \hat{\phi}_{i+1} = \sum_{n=0}^N \frac{a_n}{2} (\phi_{i+n} + \phi_{i-n}) \quad (17)$$

Here, ϕ_{i+n} and ϕ_{i-n} denote the unfiltered values of ϕ at $i+n$ and $i-n$ respectively, while $\hat{\phi}_{i-1}$, $\hat{\phi}_i$, and $\hat{\phi}_{i+1}$ are the filtered values obtained by solving the resulting tridiagonal system. The filter coefficients are determined based on the filter order as given by Visbal and Gaitonde (2002) [6].

Due to the numerical stiffness and near-wall singularity of the $k - \omega$ equations, second-order filters were required on the turbulence variables to maintain solver stability. For the remaining equations, higher-order filters were employed; however, it was observed that filter orders of eight and above provided insufficient dissipation for turbulent fluctuations to develop; filters up to sixth order were therefore used. The filter order was set at least two orders higher than the discretisation order.

Second-order and fourth-order schemes were evaluated to assess computational efficiency. The fourth-order explicit scheme with sixth-order filtering (E4F6) was found to be approximately twice as fast as the fourth-order compact scheme with the same filter order (C4F6), while achieving comparable accuracy. E4F6 was therefore selected for all subsequent computations, with boundary points discretised using the same explicit fourth-order scheme.

2.4 Pseudo Time-Stepping

Pseudo time-stepping is employed for steady-state simulations, where the unsteady equations are marched in pseudo time until convergence is achieved. Convergence is accelerated by adapting the time-step locally at each grid point based on the local flow conditions for a fixed CFL number. The local time-step is given by:

$$\Delta t = \frac{CFL}{\max \left(|U| + c\sqrt{\xi_{x_i}\xi_{x_i}}, |V| + c\sqrt{\eta_{x_i}\eta_{x_i}}, |W| + c\sqrt{\zeta_{x_i}\zeta_{x_i}} \right)} \quad (18)$$

where U , V , and W are the contravariant velocities and c is the local speed of sound.

3 RESULTS AND DISCUSSION

3.1 Validation of the $k-\omega$ Model

Firstly, a standard $k-\omega$ solver was implemented, and the results obtained were validated against the reference data of NASA's compressible RANS solver CFL3D.

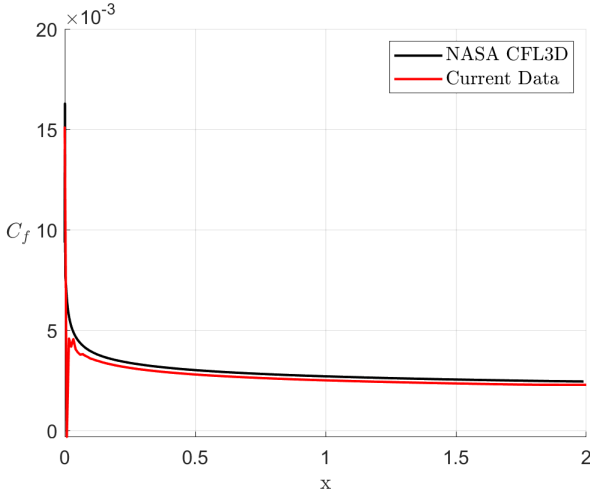


FIGURE 2: C_f versus x

Figure 2 shows the skin friction coefficient C_f distribution along the flat plate. A deviation is observed near $x = 0$, corresponding to the leading edge of the plate, where the flow transitions from the symmetry boundary condition to the wall boundary condition. This error is attributed to the coarseness of the grid near the leading edge and is expected to diminish with grid refinement. Moreover, the converged value, although very close to the CFL3D data, is seen to under-predict C_f , the error is around 7%.

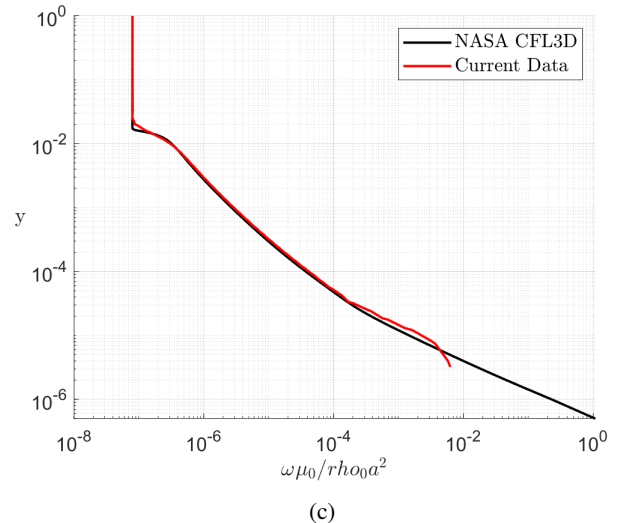
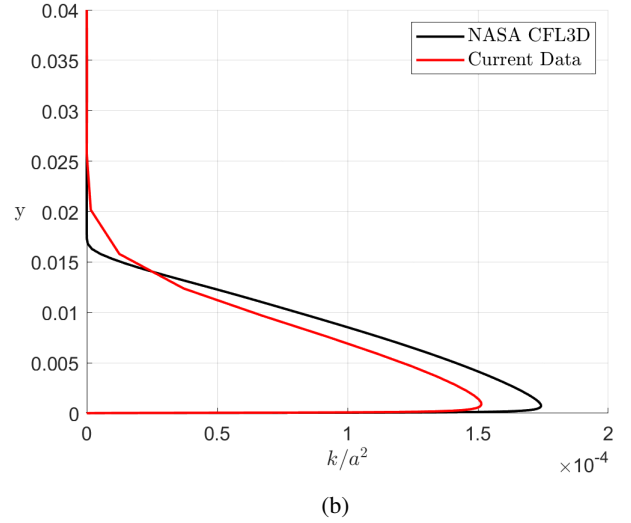
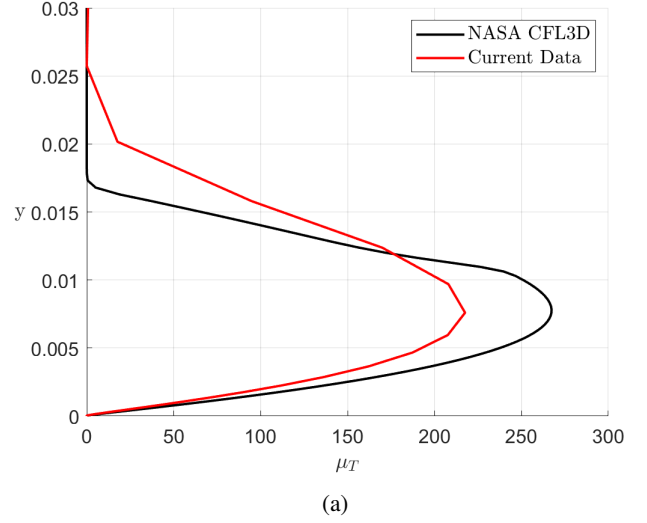
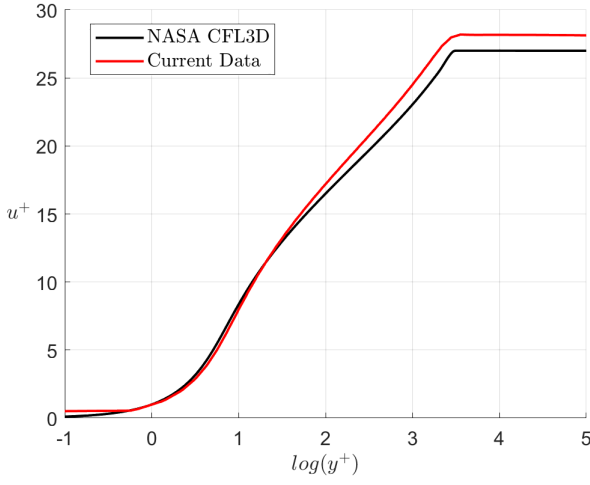
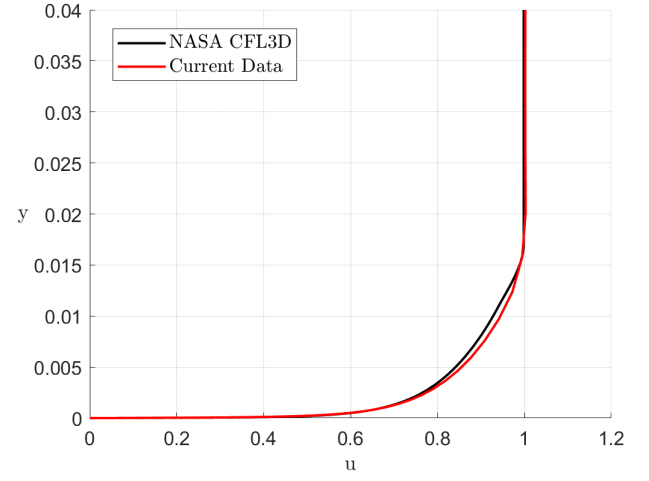


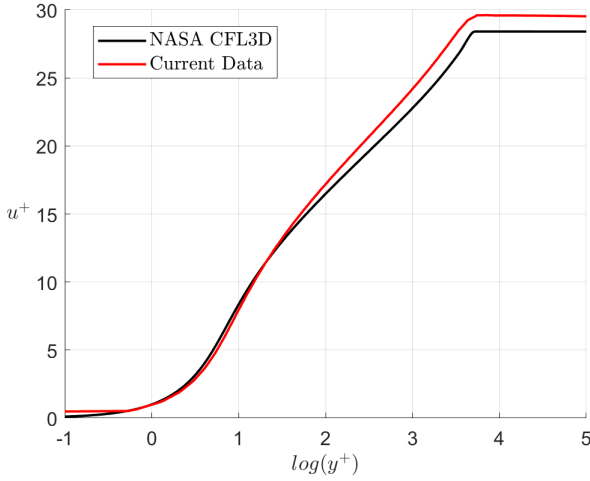
FIGURE 3: Variation of (a) μ_T (b) k (c) ω with y at $x = 0.97$



(a) $x = 0.97$

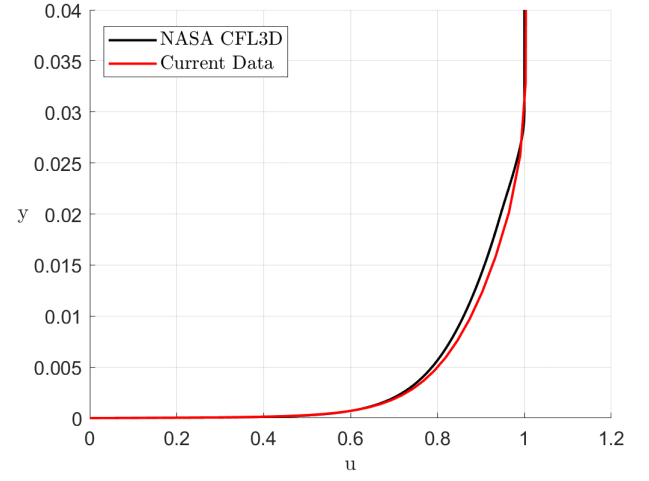


(a) $x = 0.97$



(b) $x = 1.90$

FIGURE 4: Variation of u^+ with $\log(y^+)$



(b) $x = 1.90$

FIGURE 5: Variation of u with y

Figure 3 shows deviations in the predictions of the turbulent kinetic energy and turbulent eddy viscosity. However, the current data agrees very closely to CFL3D in terms of the specific dissipation, ω . Figures 4 and 5 show the variation of u^+ with $\log(y^+)$ and u with y at two streamwise locations, $x = 0.97$ and $x = 1.90$. The velocity profiles show good agreement with CFL3D at both locations, correctly reproducing the log-law region. A small overprediction is observed in the wake region.

Overall, the present implementation of the standard $k - \omega$ model shows reasonable agreement with the NASA CFL3D reference data across the key flow quantities examined.

Grid Convergence Study:

A grid convergence study was performed to assess the sensitivity of the solution to grid refinement.

Code	p	e_a^{21}	e_{ext}^{21}	GCI_{fine}^{21}
CFL3D	1.151	1.491%	1.247%	1.527%
Current	0.928	2.568%	2.810%	3.618%

TABLE 1: Grid Convergence Index (GCI) Results

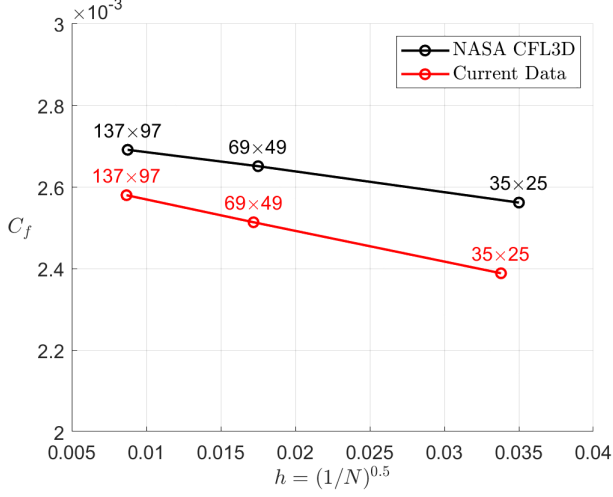


FIGURE 6: Variation of C_f with grid size at $x = 0.97$

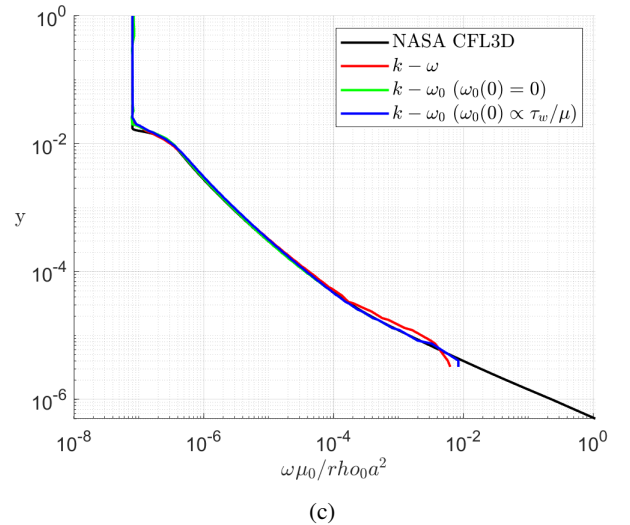
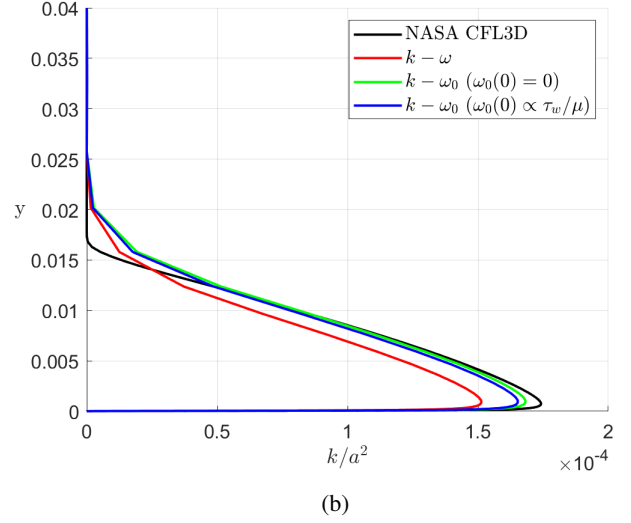
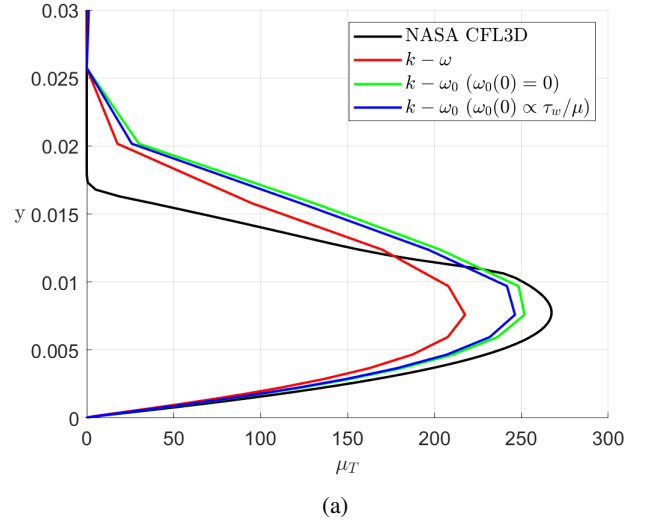


FIGURE 8: Variation of (a) μ_T (b) k (c) ω with y at $x = 0.97$

3.2 Results of the $k-\omega_0$ Model

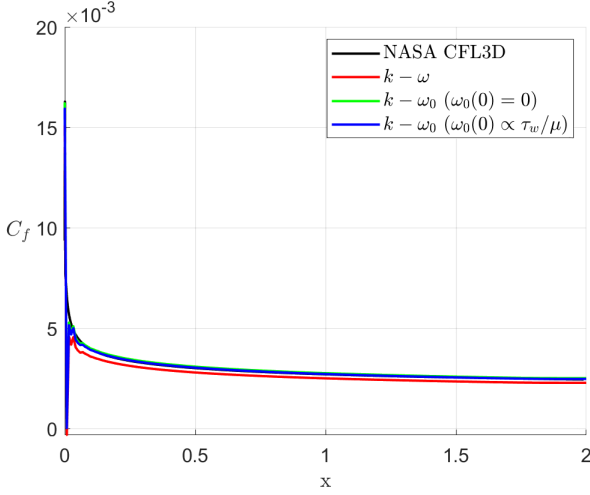
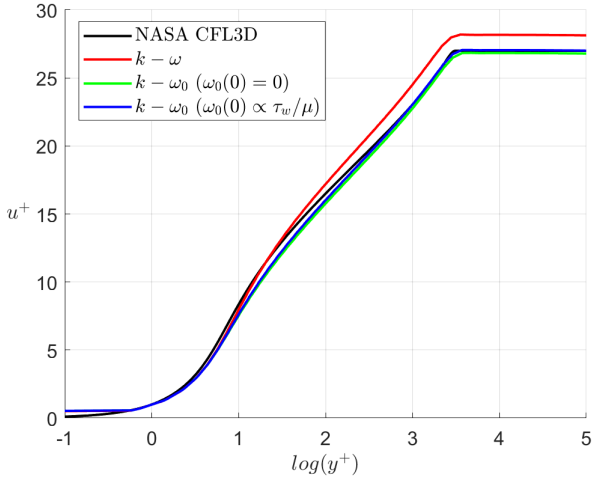


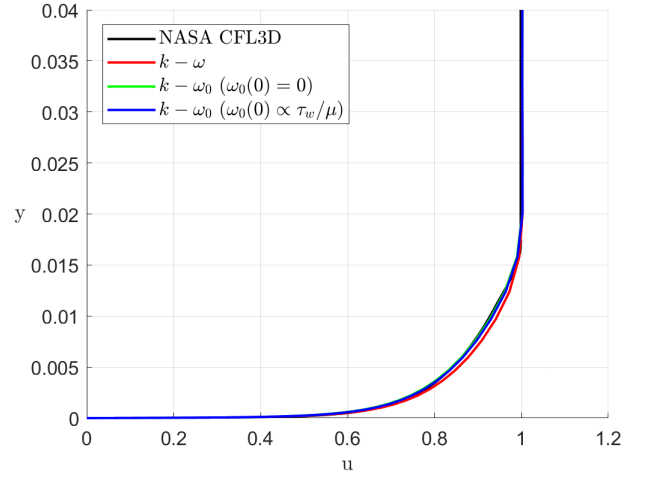
FIGURE 7: C_f versus x

Figure 7 shows the skin friction coefficient distribution for the standard $k-\omega$ model and the two $k-\omega_0$ formulations against the CFL3D reference data. The $k-\omega_0$ formulation show a noticeable improvement over the standard $k-\omega$ model, with both boundary conditions producing results closer to the CFL3D reference.

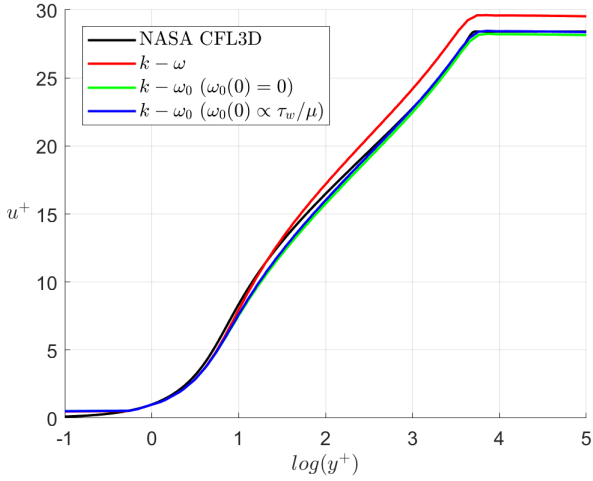
Figure 8 shows that the $k-\omega_0$ formulation more accurately captures the peak values and overall profiles of all three quantities as compared to the standard $k-\omega$ model. The $\omega_0(0) = 0$ boundary condition seems to be slightly more accurate than the other.



(a) $x = 0.97$

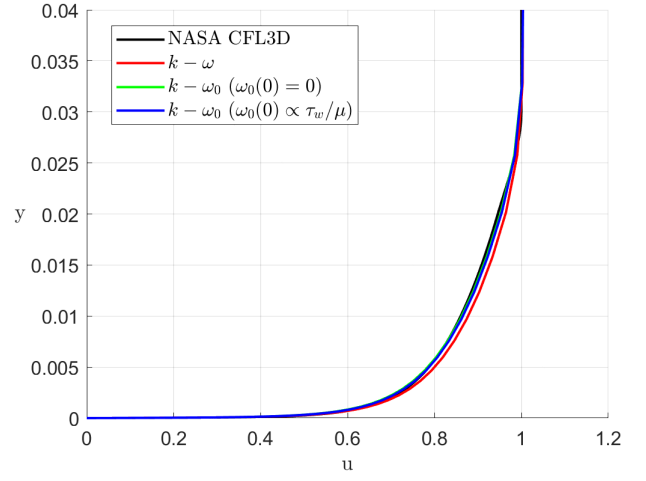


(a) $x = 0.97$



(b) $x = 1.90$

FIGURE 9: Variation of u^+ with $\log(y^+)$

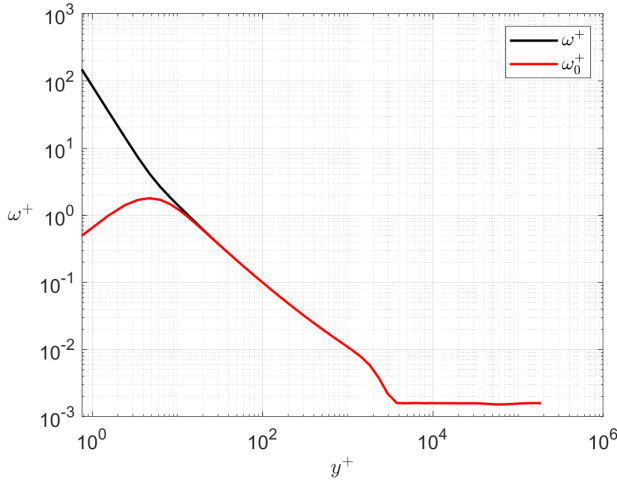


(b) $x = 1.90$

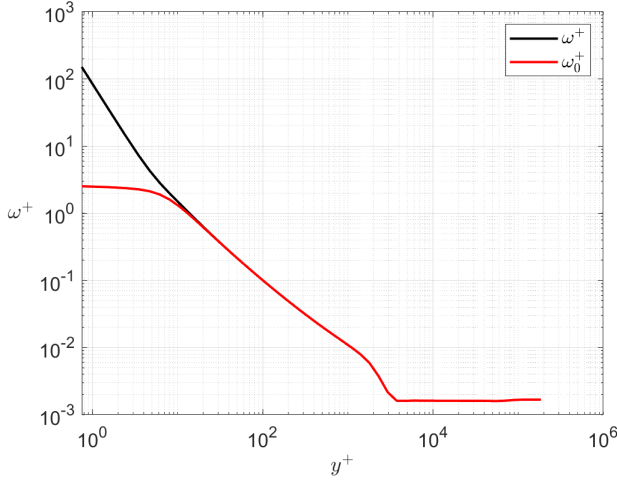
FIGURE 10: Variation of u with y

Figures 9 and 10 show the variation of u^+ with $\log(y^+)$ and u with y at two streamwise locations, $x = 0.97$ and $x = 1.90$. The $k - \omega_0$ formulations are in closer agreement with the CFL3D reference than the standard $k - \omega$ model at both locations. The improvement is most evident in the outer region of the boundary layer, where the standard $k - \omega$ model shows a slight overprediction of u^+ . Both $k - \omega_0$ boundary conditions produce nearly identical velocity profiles, indicating insensitivity of the solution to the choice of wall boundary condition. This demonstrates that the reformulation improves predictive accuracy without any major modifications to the underlying turbulence model equations.

Figure 11 shows the comparison of ω^+ and ω_0^+ for both boundary conditions. ω_0^+ remains regular and bounded at the wall. The two solutions converge beyond $y^+ \approx 10$, consistent with the findings of Durbin and Yin [3], confirming that the transformation is active only in the near-wall region. For the first boundary condition, $\omega_0(0) = 0$, ω_0^+ rises sharply from zero before converging to ω^+ . The second boundary condition, $\omega_0(0) = \sqrt{\alpha/\beta_0} \tau_w/\mu$, results in ω_0^+ being nearly constant in the viscous sublayer, avoiding the sharp near-wall gradient entirely. This makes the second boundary condition numerically more attractive, as it reduces the stiffness of the system in the near-wall region.



(a) $\omega_0(0) = 0$



(b) $\omega_0(0) = \sqrt{\frac{\alpha}{\beta_0} \frac{\tau_w}{\mu}}$

FIGURE 11: Comparison of ω to ω_0

From the figures above, it can be observed that the $k - \omega_0$ model provides more accurate results than the $k - \omega$ model, on the same grid with identical solver settings.

Numerical Performance:

1. The standard $k - \omega$ formulation required second-order filters on both the k and ω equations to maintain solver stability, owing to the near-wall singularity.
2. The $k - \omega_0$ formulation required no filtering on the k and ω_0 equations; the standard sixth-order filter applied to the remaining flow variables was sufficient.

3. The ability to run without low-order filters on the turbulence equations is a direct indicator of reduced numerical stiffness, as the regular wall behaviour of ω_0 removes the destabilising effect of the singularity.
4. A wall-clock time reduction of at least 10% was observed across all $k - \omega_0$ configurations compared to the standard $k - \omega$ formulation, consistent with the reduced numerical stiffness of the reformulated equations.

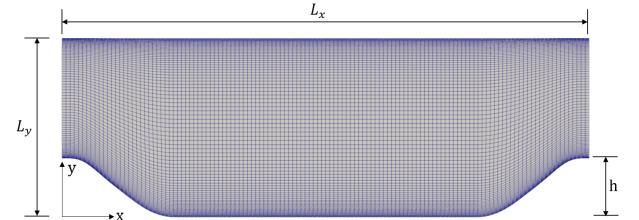
3.3 Application to Periodic Hill

Geometry and Flow Features

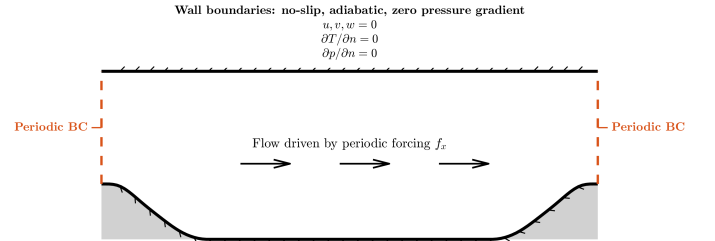
The periodic-hill geometry follows the standard benchmark used by Frohlic et al [7]. The domain length, channel height, and span nondimensionalized by the hill height h are

$$L_x/h = 9, \quad L_y/h = 3.035, \quad L_z/h = 4.5. \quad (19)$$

where h is 28.0mm. The lower wall consists of a smooth periodic hill, and the upper wall is a flat solid boundary. The stream and span directions are periodic. As the flow passes through the crest of the hill, it separates approximately $0.2h$ downstream of the crest and forms a large recirculation bubble. The flow reattaches around $4.6h - 4.7h$. Following reattachment, the flow accelerates up the windward face of the subsequent hill. The geometry is shown in Figure 12a. Figure ?? shows the boundary conditions of the geometry considered.



(a) A two-dimensional slice of the geometry considered



(b) Boundary condition of the geometry

FIGURE 12: Computational domain and boundary conditions.

Periodic Hill Grid Generation

Hill Profile. The hill profile is built using piecewise polynomial that dictate the descending face from the crest down to the flat wall, depending upon where x falls within specific intervals. The reference piecewise cubic polynomials is taken from [8]. The curved hill profile occupies $0 \leq x/h \leq 54/28$ at the upstream side and is mirrored over to $7.0714 \leq x/h \leq 9 - 54/28$

Grid Clustering. A hyperbolic tangent stretching function is used to heavily cluster the grid nodes in the wall normal direction near both the upper and lower boundaries.

Orthogonality and blending. Near the curved lower wall, the first few grid lines were aligned approximately with the normal direction of the local inward wall to improve orthogonality. A smooth blending function was used to transition these near wall normal lines into nearly vertical grid lines toward the upper wall.

Results

Figure 13 shows the lower wall skin friction coefficient plotted against the lower wall as function of x/h . The comparison is made between the standard $k - \omega$ model and $k - \omega_0$ formulation. The simulation was carried out at a Reynolds number of $Re = 10595$ based on the hill height and bulk velocity above the crest. The computational grid used was $129 \times 97 \times 3$. The governing equations were solved using a fourth order explicit scheme in space and a four stage Runge-Kutta scheme in pseudo time. Local time stepping was employed to accelerate convergence with a CFL of 0.1. Second order filter with $\alpha_f = 0.45$ was applied to both the mean flow and turbulence flow variables. No slip conditions were imposed on both the lower hill and upper wall, while streamwise and spanwise periodicity were imposed. The bulk velocity was maintained at 1 unit.

Both the model predict nearly identical skin-friction distribution throughout the domain. This behaviour is consistent with the objective of the Durbin-Yin [3] reformulation, which is to remove the numerical stiffness associated with standard model while retaining the same physical behaviour as the original model.

At $x/h = 1.0$, the flow separates. Between approximately $x/h = 0.1$ and $x/h = 3.4$, C_f becomes negative. This negative skin friction region corresponds to reverse wall shear and represents the recirculation bubble. Reattachment occurs at $x/h = 3.4$ and downstream of this point, C_f becomes positive again, indicating the reattachment of the boundary layer to the lower wall

Although separation and reattachment are clearly captured, the computed bubble is shorter than that reported in the reference LES studies of Fröhlich et al. [7]. In the benchmark data, separation

occurs much earlier, around $x/h = 0.2$, and reattachment around $x/h = 4.6 - 4.7$. In the present simulation, the separation occurs much later and reattachment occurs earlier. This suggests that the models are qualitatively correct but they don't match the benchmark. Furthermore, the dominant source of error is not the $k - \omega_0$ reformulation itself, but could be attributed to the grid resolution, diffusion or forcing methodology.

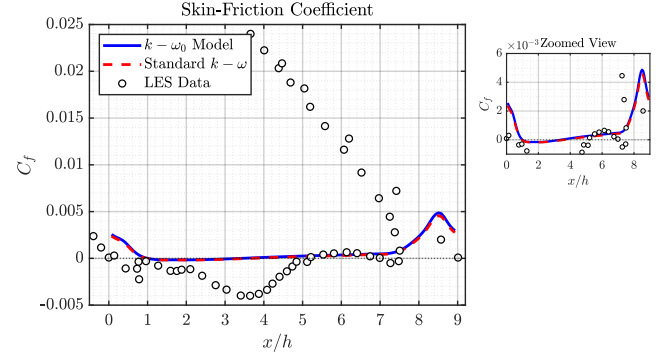
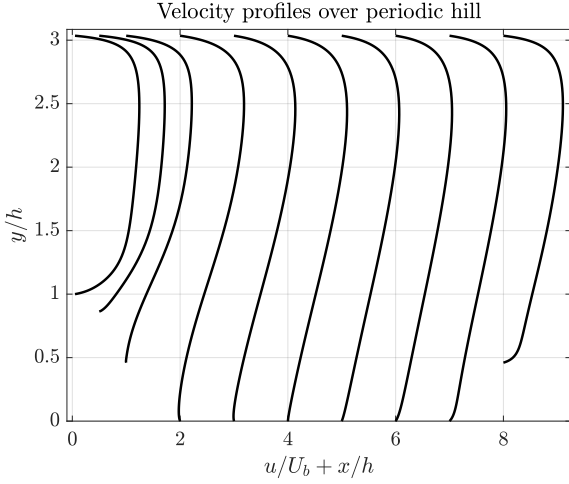


FIGURE 13: Comparison of lower wall skin-friction coefficient distributions predicted by the Wilcox(2006) $k - \omega$ model and the $k - \omega_0$ model against benchmark for flow over a periodic hill at $Re_h = 10595$ with $\omega_0(0) = \sqrt{\frac{\alpha}{\beta_0}} \frac{\tau_w}{\mu}$ used as the boundary condition for ω_0 . Negative C_f values indicate region of flow recirculation. The two formulations produce almost identical predictions of separation and reattachment

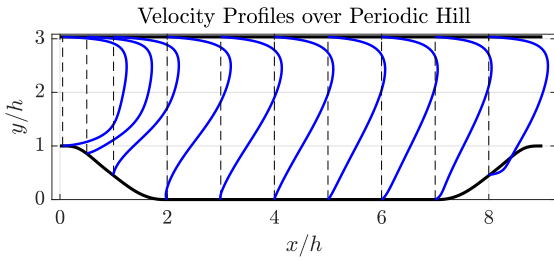
Figure 14 shows the development of streamwise velocity over the periodic hill. Downstream of the crest, the near wall velocity decreases strongly due to the adverse pressure gradient. The slight leftward bending of the lower part of the profiles indicates the separated region. Further downstream, the profiles gradually recover as the boundary layer reattaches to the wall. Near the end of the domain, showing strong acceleration toward the next hill crest. The bending near the top boundary occurs because of the no-slip wall.

3.4 Extension to Higher Mach Numbers

The $k - \omega_0$ formulation is constructed such that the transformation is consistent with local wall scaling in compressible flow, providing a physically more accurate near-wall boundary condition than the standard $k - \omega$ model. This motivates the extension of the present solver to higher Mach numbers, where compressibility effects become significant and the near-wall treatment of ω has a greater influence on the solution. The solver was modified to account for compressibility effects. A WENO scheme was employed for discretisation in the x -direction to



(a)

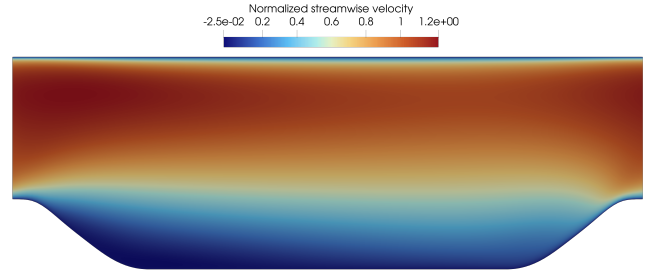


(b)

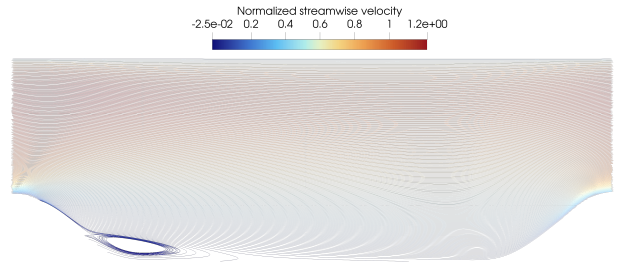
FIGURE 14: Normalized streamwise velocity profiles over the periodic hill at selected streamwise locations. The profiles are shifted by their corresponding x/h positions.

suppress spurious oscillations near shocks, which are characteristic of central difference schemes at supersonic conditions.

The two formulations were compared for flat plate boundary layer flow at Mach 1.2 and a Reynolds number of 5×10^6 . The velocity profiles in figure 16 show a noticeable difference between the two formulations. The standard $k - \omega$ model overpredicts u^+ in the log-law and outer regions, while the $k - \omega_0$ formulation produces a lower value. Figure 17 shows the density contour for the Mach 1.2 case, where a weak oblique shock emanating from the leading edge of the plate is clearly visible.



(a) Contour plot of the normalized streamwise velocity.



(b) Streamlines depicting the recirculation zone.

FIGURE 15: Flow field visualization over the periodic hill.

4 CONCLUSIONS

The present study successfully implemented and validated both the standard $k - \omega$ and the reformulated $k - \omega_0$ turbulence model using a compressible Reynolds-Averaged Navier-Stokes (RANS) framework.

For the flat plate boundary layer validation, the $k - \omega_0$ formulation yielded skin friction and velocity profiles that match the NASA CFL3D reference data more closely than the standard model. Moreover, the transformation to the regular ω_0 variable alleviated the numerical stiffness caused by the standard model's near-wall singularity. This reduction in stiffness allowed the $k - \omega_0$ solver to remain stable without the need for low-order filters on the turbulence transport equation, which were strictly required for the standard model. Additionally the improved numerical performance, reduced the wall clock time by at least 10% across all $k - \omega_0$ configurations. Testing of both the boundary conditions revealed that setting $\omega_0(0) \propto \tau_w/\mu$ is particularly advantageous, as it keeps the specific dissipation rate nearly constant near the wall.

When applied to the periodic hill test case, both the models

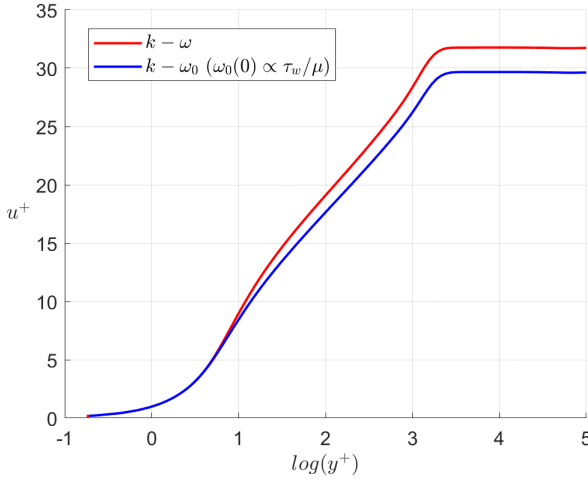


FIGURE 16: Variation of u^+ with $\log(y^+)$ at $M = 1.2$ and $x = 0.97$

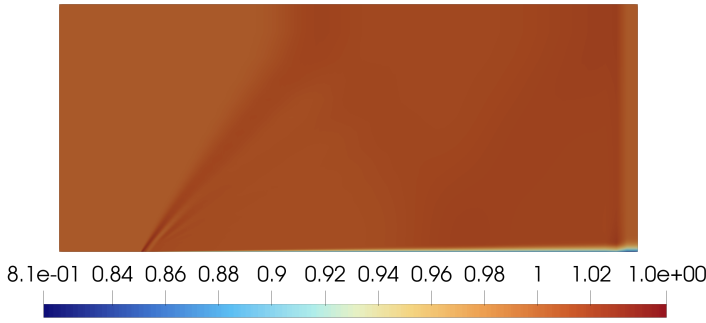


FIGURE 17: Density Contour, $M = 1.2$

predicted nearly identical skin-friction distributions and recirculation zones. This confirms that the $k - \omega_0$ does give numerical robustness and stability while retaining the physical behaviour of the standard model. However, the predicted recirculation bubble in the periodic hill was the reference LES benchmark data. This deviation was attributed to factors such as grid resolution, numerical diffusion or forcing methodology rather than the turbulence model formulation itself.

The models were used to simulate the boundary layer flow at Mach 1.2, where the u^+ versus $\log(y^+)$ plot revealed that the $k - \omega$ model overpredicts the u^+ value as compared to $k - \omega_0$.

FUTURE SCOPE

Several avenues remain open for further investigation. The present solver treats each test case independently; future work

could focus on integrating multiple flow configurations into a unified solver framework. The extension of the $k - \omega_0$ formulation to hybrid RANS/LES approaches such as Detached Eddy Simulation (DES) and Delayed Detached Eddy Simulation (DDES) would enable the capture of a wider range of flow physics. Finally, a more systematic study of the observed reduction in computational time is required, as the precise mechanisms behind the improved convergence behaviour of the $k - \omega_0$ formulation are not yet fully understood.

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